SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I saravanan.T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Saravanan.t

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x$: Vehicle(x) $\wedge$ EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]

II. $\forall x$: ClearPath(x) $\rightarrow$ GreenSignal(x) $\wedge \neg$RedSignal(x) [Rule 2]

III. $\forall x \forall y$: Vehicle(x) $\wedge$ Behind(x,y) $\wedge$ GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

LOGICAL AGENTS — CASE STUDIES 1 TO 4

Complete assignment answers with propositional logic, FOL, inference rules, CNF, resolution, and reasoning trees.

CASE STUDY 1 — SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic representation

Symbols: F = Fever; C = Cough; R = Rash; B = Breathlessness; FL = Possible Flu; M = Possible Measles; P = Possible Pneumonia.

Logical operators: ∧ means AND; → means implies; ¬ means NOT.

$$(F \land C) \rightarrow FL$$

$$(F \land R) \rightarrow M$$

$$(C \land B) \rightarrow P$$

Facts for Patient A: F = True, C = True, B = True. Rash is not given as true.

(b) Forward Chaining

Forward Chaining starts with known facts and repeatedly fires rules whose premises are satisfied.

• Initial facts: F, C, B.

• Rule 1: (F ∧ C) → FL. Both F and C are true, so derive FL (Possible Flu).

• Rule 2: (F ∧ R) → M. R is not known to be true, so M (Possible Measles) cannot be derived.

• Rule 3: (C ∧ B) → P. Both C and B are true, so derive P (Possible Pneumonia).

Result: Possible Flu and Possible Pneumonia are derived. Possible Measles is not derivable from the supplied facts.

(c) Backward Chaining and goal reduction tree

Goal: P. Rule 3 says (C ∧ B) → P, so reduce the goal to C and B. Both are facts, therefore the goal succeeds.

Goal: P (Possible Pneumonia)	
Rule 3: (C ∧ B) → P	
Subgoal C Cough = True SUCCESS	**Subgoal B** Breathlessness = True SUCCESS
Conclusion: P is proved	

CASE STUDY 2 — AUTONOMOUS TRAFFIC MANAGEMENT AGENT

(a) Unification of Rule 1

Rule 1: ∀x [Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)]. Facts: Vehicle(Ambulance), EmergencyType(Ambulance).

Step 1: Vehicle(x) matches Vehicle(Ambulance), so θ■ = {x/Ambulance}.

Step 2: EmergencyType(x) matches EmergencyType(Ambulance), consistent with θ■ = {x/Ambulance}.

$$MGU: \theta = \{x/Ambulance\}$$

Substitute x = Ambulance: (Vehicle(Ambulance) ∧ EmergencyType(Ambulance)) → ClearPath(Ambulance). Thus derive ClearPath(Ambulance).

(b) Resolution proof of Proceed(CarA)

Convert implications to CNF using A → B ≡ ¬A ∨ B and distribute ∨ over ∧.

$$C1: \neg Vehicle(x) \lor \neg EmergencyType(x) \lor ClearPath(x)$$

$$C2: \neg ClearPath(x) \lor GreenSignal(x)$$

$$C3: \neg ClearPath(x) \lor \neg RedSignal(x)$$

$$C4: \neg Vehicle(x) \lor \neg Behind(x,y) \lor \neg GreenSignal(y) \lor Proceed(x)$$

Facts: C5 Vehicle(Ambulance); C6 EmergencyType(Ambulance); C7 Vehicle(CarA); C8 Behind(CarA,Ambulance). Add the relevant ground instances and derive ClearPath(Ambulance) from C1, C5, C6, then GreenSignal(Ambulance) from C2 and

ClearPath(Ambulance).

Resolution refutation: negate the goal, C9 = ¬Proceed(CarA). Instantiate C4 with x = CarA, y = Ambulance.

Step	Resolve	Result
1	C4[x=CarA,y=Ambulance] with C9	¬Vehicle(CarA) ∨ ¬Behind(CarA,Ambulance) ∨ ¬GreenSigna
2	Resolve with C7: Vehicle(CarA)	¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance)
3	Resolve with C8: Behind(CarA,Ambulance)	¬GreenSignal(Ambulance)
4	Resolve with derived GreenSignal(Ambulance)	■ (empty clause)

Conclusion: The negated goal produces the empty clause ■, so Proceed(CarA) follows from the stated KB.

(c) Two simultaneous emergency vehicles

The current KB is not sufficient to safely coordinate two simultaneous emergencies. It can derive ClearPath and GreenSignal for each emergency vehicle independently, but it does not represent route conflicts, junction occupancy, priorities, or signal timing.

• Add facts identifying each emergency vehicle, for example Vehicle(Ambulance1), EmergencyType(Ambulance1), Vehicle(FireTruck1), EmergencyType(FireTruck1).

• Add Route(vehicle, road), At(vehicle, junction), and Conflict(route1, route2) predicates to represent route overlap.

• Add priority/scheduling rules, such as HigherPriority(x,y), and a rule that lets the higher-priority vehicle pass first when routes conflict.

• Add signal-safety constraints preventing conflicting movements from receiving green at the same time, and timing/clearance facts to ensure one vehicle clears the junction before the other proceeds.

Justification: Without conflict, priority, and timing information, the KB cannot determine whether both vehicles may safely proceed at once.

CASE STUDY 3 — AGRICULTURAL EXPERT REASONING SYSTEM

(a) Convert P1, P2, P3 and facts to CNF

Let S = SoilDry; I = IrrigationNeeded; W = CropWheat; D = ApplyDripMethod; A = CropAtRisk.

P1: S → I. Eliminate implication: ¬S ∨ I. This is already CNF.

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P1 CNF: ¬S ∨ I
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P2: I ∧ W → D. Eliminate implication: ¬(I ∧ W) ∨ D. Apply De Morgan's law: ¬I ∨ ¬W ∨ D. This is CNF.

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P2 CNF: ¬I ∨ ¬W ∨ D
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P3: ¬D ∧ W → A. Eliminate implication: ¬(¬D ∧ W) ∨ A. Apply De Morgan's law: D ∨ ¬W ∨ A. This is CNF.

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P3 CNF: D ∨ ¬W ∨ A
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Facts S = True and W = True become unit clauses:

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S ; W
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(b) Resolution refutation to prove ApplyDripMethod

Goal: D. Negate the goal and add ¬D to the CNF knowledge base.

• C1: ¬S ∨ I

• C2: ¬I ∨ ¬W ∨ D

• C3: D ∨ ¬W ∨ A

• C4: S; C5: W; C6: ¬D (negated goal)

Resolution steps:

• Resolve C1 (¬S ∨ I) with C4 (S) to derive I.

• Resolve C2 (¬I ∨ ¬W ∨ D) with I to derive ¬W ∨ D.

• Resolve ¬W ∨ D with C5 (W) to derive D.

• Resolve D with C6 (¬D) to derive ■, the empty clause.

Therefore D is proved by resolution refutation.

Proof tree (text form):

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S, (¬S ∨ I) ■ I
I, (¬I ∨ ¬W ∨ D) ■ (¬W ∨ D)
(¬W ∨ D), W ■ D
D, ¬D ■ ■
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(c) Remove SoilDry from the KB

Without S, P1 cannot derive I because its antecedent is missing. Since I is not otherwise a fact, P2 cannot derive D. However, P3 alone does not establish CropAtRisk: it requires both ¬D and W.

Under the open-world assumption used in ordinary logic, failure to derive D does not mean ¬D is true. Therefore CropAtRisk does not follow from the remaining KB unless ¬D is explicitly added as a fact or established by another rule.

If ¬D is explicitly given and W remains true, P3 derives A (CropAtRisk). Without that additional fact, CropAtRisk is not provable.

CASE STUDY 4 — WUMPUS WORLD KNOWLEDGE AGENT

(a) Belief state and Modus Ponens

Percepts: Stench(1,2), Breeze(1,1), Glitter(2,2).

Rules: Stench(1,2) → Wumpus is adjacent to (1,2); Breeze(1,1) → Pit is adjacent to (1,1); Glitter(x,y) → Gold(x,y); ¬Wumpus(x,y) ∧ ¬Pit(x,y) → Safe(x,y).

• From Stench(1,2), by Modus Ponens, infer that at least one cell adjacent to [1,2] contains the Wumpus.

• From Breeze(1,1), infer that at least one cell adjacent to [1,1] contains a pit.

• From Glitter(2,2), instantiate the gold rule and derive Gold(2,2).

The supplied rules do not identify one exact Wumpus or pit location, and they do not establish that any specific cell is safe. Rule 4 can derive Safe(x,y) only when both ¬Wumpus(x,y) and ¬Pit(x,y) are known.

(b) Forward Chaining: possible locations

Assuming the standard four-neighbour grid and that the agent is inside the grid:

• Wumpus candidates from Stench at [1,2]: adjacent cells [1,1], [1,3], and [2,2] (and [0,2] only if coordinates outside the usual positive grid are allowed).

• Pit candidates from Breeze at [1,1]: adjacent cells [1,2] and [2,1] (and [0,1], [1,0] only if such cells are allowed).

These are candidate sets, not definite locations. The given KB has no rule that excludes candidates based on lack of a percept, so more facts/percepts are needed to narrow them.

(c) Is path [1,1] → [1,2] → [2,2] safe?

The KB does not prove this path safe. Breeze at [1,1] makes [1,2] a possible pit location, and Stench at [1,2] makes nearby cells possible Wumpus locations. Glitter confirms gold at [2,2], but does not prove that [2,2] is free of hazards.

To infer Safe(cell), the KB requires both ¬Wumpus(cell) and ¬Pit(cell). Those negative facts are not supplied for [1,1], [1,2], or [2,2]. Therefore, the agent cannot logically certify the entire path as safe from the given information.

Conclusion: Gold is known to be at [2,2], but the path's safety remains undetermined. The agent should gather more percepts or use additional hazard-exclusion rules before treating the route as safe.

Important logical note: In Wumpus World, 'possible adjacent hazard' does not mean the hazard's exact location is known. Safety requires explicit evidence that a cell contains neither a Wumpus nor a pit.